Forecasting and Macroeconomic Analytics

ECON 522

Elena Pesavento — Emory University

Spring 2025

Trends

Trend

- Trend is a slow, long run evolution in the variable that we want to model
- There are two kind of trends: *Deterministic* and *Stochastic*
- It is also called **deterministic trend** because evolves in predictable way. How do we model a determinisitc trend?

$$y_t = c + \beta t + u_t$$

- **Stochastic trends** are non predictable (e.g unit roots, random walk)

Deterministic Trend

There are various kinds of deterministic trends:

- Linear Trend

$$y_t = c + \beta t \rightarrow$$

no error terms (yet)

- Quadratic Trend

$$y_t = c + \beta_1 t + \beta_2 t^2$$

- Exponential Trend

$$y_t = ce^{\beta_1 t} \text{ or } \log(y_t) = \log(c) + \beta_1 t$$

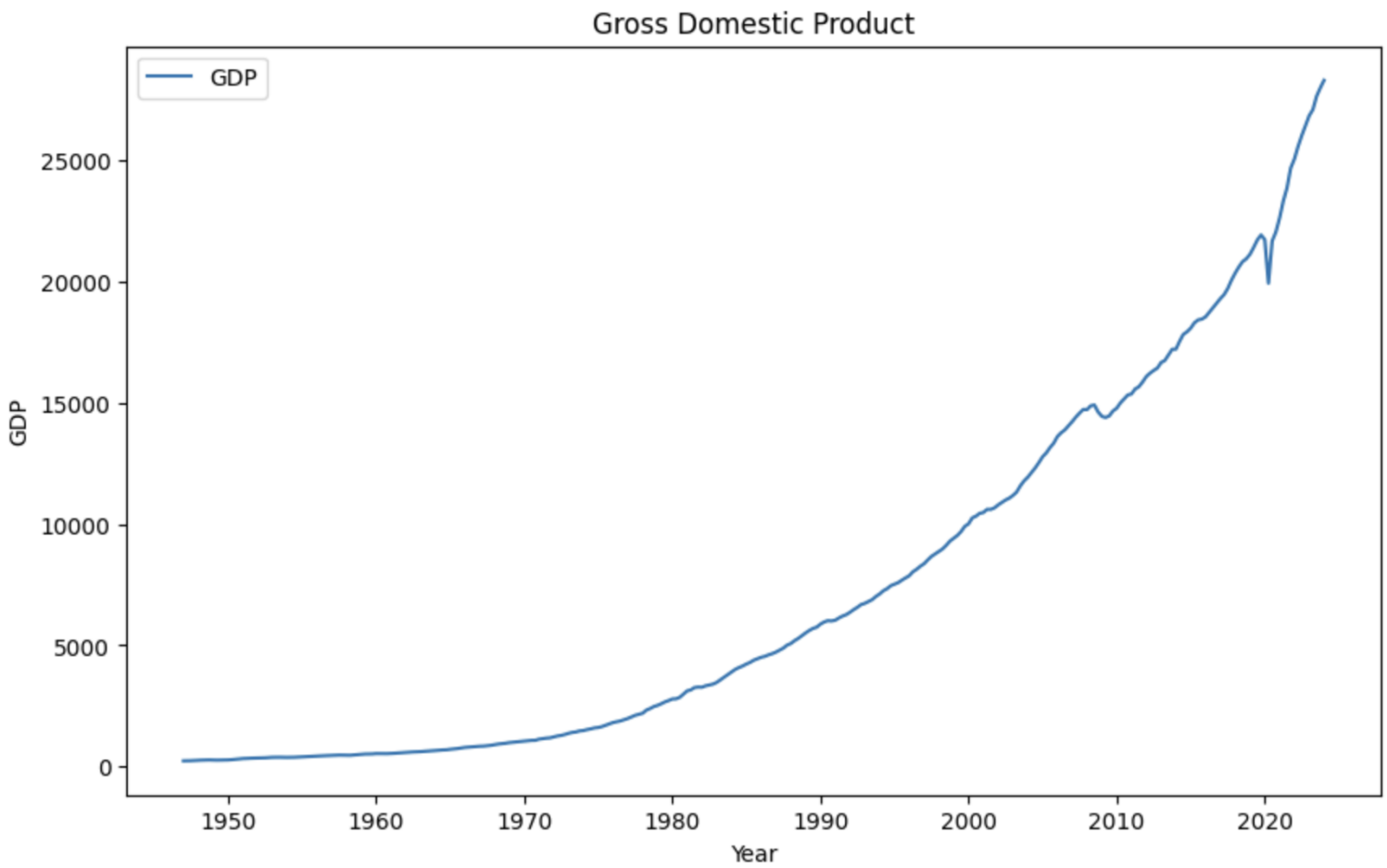
There are other kinds of trends but those three approximate any trend fairly well!

> Adding an explanatory variable that looks like a trend will explain the same property.

When you add explanatory variable that also goes up and follows trend there will be a lot of Correlation so you have to be careful

Looking Closer at GDP

Follow the Module 1 Python code file to graph this using the data 'GDP.csv'



How do we estimate a Deterministic Trend?

The model we have in mind is

$$GDP_t = c + \beta t + u_t$$

We can run this using any statistical software using some OLS command and we get the following

*constant doesn't have an interpretation, just leave it in

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.865
Model:                            OLS   Adj. R-squared:                  0.865
Method:                 Least Squares   F-statistic:                     1973.
Date:                Tue, 23 Jul 2024   Prob (F-statistic):          1.03e-135
Time:                        19:45:10   Log-Likelihood:                -2884.3
No. Observations:                 309   AIC:                             5773.
Df Residuals:                     307   BIC:                             5780.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant   -4763.0379    311.942    -15.269      0.000   -5376.853   -4149.223
Time          77.8620      1.753     44.422      0.000      74.413      81.311
==============================================================================
Omnibus:                       34.120   Durbin-Watson:                   0.005
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               42.573
Skew:                           0.902   Prob(JB):                     5.69e-10
Kurtosis:                       3.236   Cond. No.                         355.
==============================================================================
```

This is a time series regression of GDP on a linear time trend. Let me walk through what the output reveals—and what should concern us.

The Basic Results

Time coefficient (77.86): GDP increases by about 77.86 units per time period on average. The t-stat of 44.4 is massive, and $R^2 = 0.865$ says the linear trend explains 86.5% of GDP variation.

****Constant (-4763):**** The intercept when t = 0. Depending on how time is coded, this might be extrapolating way outside your data range, so don't interpret it literally.

The Red Flags

****Durbin-Watson = 0.005:**** This is the critical diagnostic here. Durbin-Watson tests for serial correlation in residuals, and it ranges from 0 to 4, with 2 indicating no autocorrelation. A value of 0.005 is essentially zero—this indicates extreme positive serial correlation in the errors.

This connects directly to the serial correlation slide you showed earlier. The consequences are exactly what that slide warned about: OLS coefficients may still be unbiased, but your standard errors are wrong (too small), which means your t-stats are inflated and your confidence intervals are too narrow.

Jarque-Bera (42.57, p ≈ 0):** This tests for normality of residuals. The rejection suggests residuals aren't normal—though with n = 309 and asymptotic properties, this is less concerning than the serial correlation.

What's Happening Economically

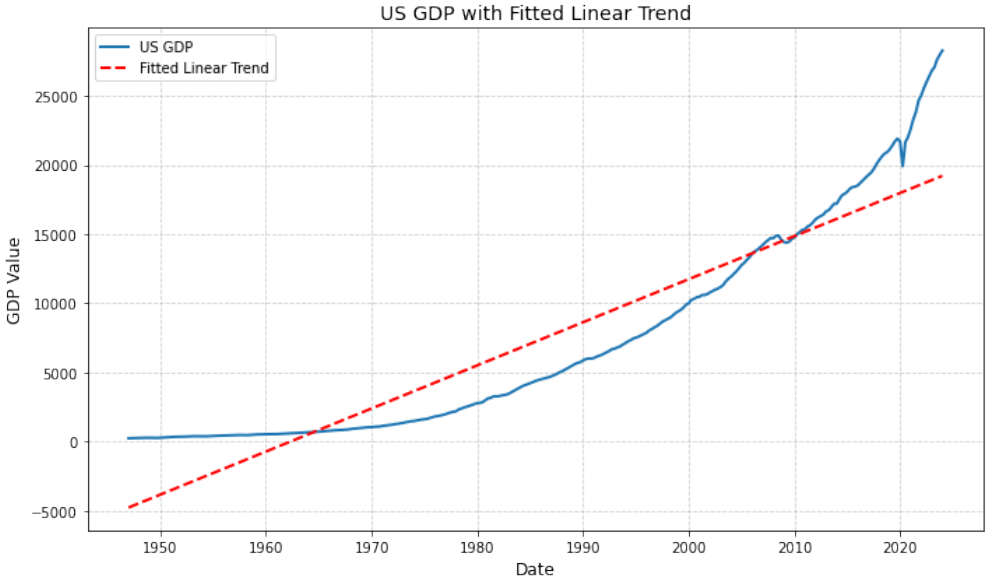
GDP doesn't just follow a smooth linear trend—it has business cycles, persistent shocks, and momentum. When GDP is above trend in period t, it's likely still above trend in period t+1. That persistence is showing up as serial correlation.

The fix: This is exactly when you'd use Newey-West HAC standard errors, or consider modeling the dynamics explicitly (AR terms, as the earlier slide mentioned).

Would you like to see how the inference would change with robust standard errors?

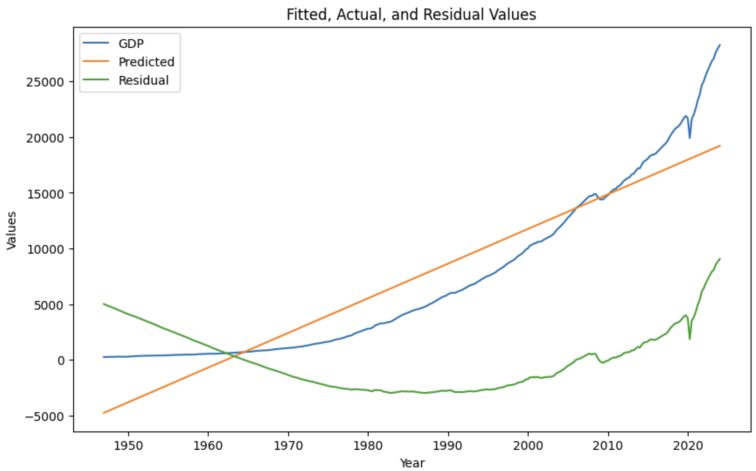
Linear Trend Model?

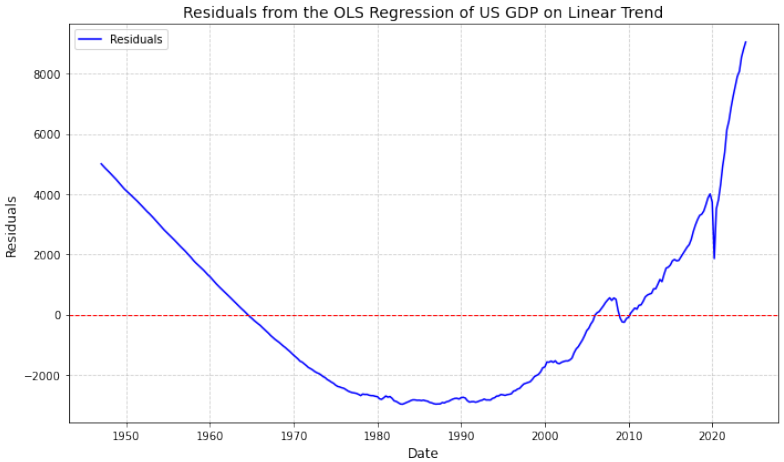
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.865
Model:                            OLS   Adj. R-squared:                  0.865
Method:                 Least Squares   F-statistic:                     1973.
Date:                Tue, 23 Jul 2024   Prob (F-statistic):          1.03e-135
Time:                        19:45:10   Log-Likelihood:                -2884.3
No. Observations:                 309   AIC:                             5773.
Df Residuals:                     307   BIC:                             5780.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant   -4763.0379    311.942    -15.269      0.000   -5376.853   -4149.223
Time          77.8620      1.753     44.422      0.000      74.413      81.311
==============================================================================
Omnibus:                       34.120   Durbin-Watson:                   0.005
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               42.573
Skew:                           0.902   Prob(JB):                     5.69e-10
Kurtosis:                       3.236   Cond. No.                         355.
==============================================================================
```



- Trend explains most of GDP variation. Typical of most time series'.
- R^2 is really high. Is this a good regression? Why or why not?
- But how do we know if this is truly deterministic trend? We will spent A LOT of time talking about this and the difference between deterministic and stochastic trends

Linear Trend Estimation?





- Look at fitted values! (What do we learn?)
- Which pattern do we want the residuals to have?

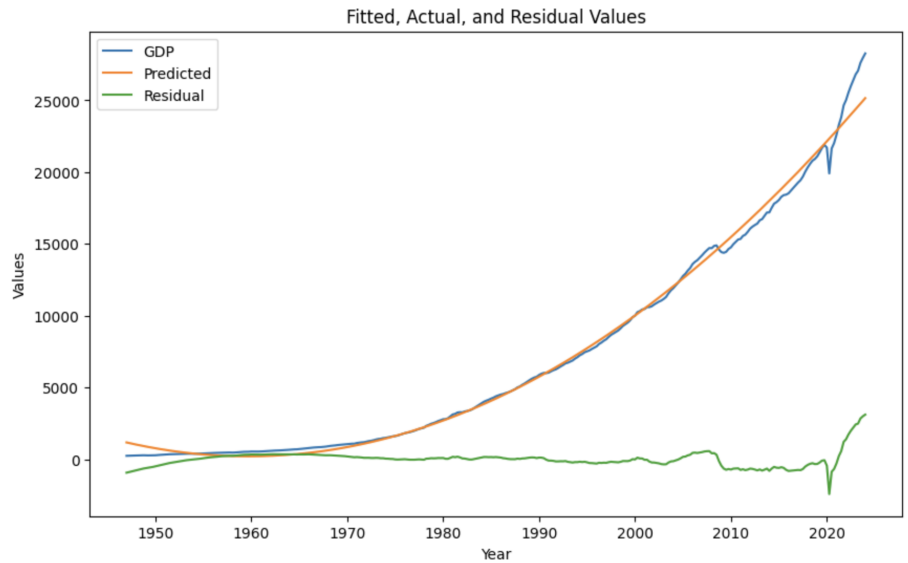
How would you expect residuals to look like if this was a good model? residual mean is zero, close to a straight model, ideally small, and random. No obvious pattern.

Quadratic Trend Model?

Let us now try a quadratic deterministic trend model!

$$GDP_t = c + \beta_1 t + \beta_2 t^2 + u_t$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.994
Model:                            OLS   Adj. R-squared:                  0.994
Method:                 Least Squares   F-statistic:                 2.683e+04
Date:                Tue, 23 Jul 2024   Prob (F-statistic):               0.00
Time:                        20:19:05   Log-Likelihood:                -2395.0
No. Observations:                 309   AIC:                             4796.
Df Residuals:                     306   BIC:                             4807.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant    1174.2928     95.795     12.258      0.000     985.793    1362.792
Time         -38.1771      1.437    -26.570      0.000     -41.004     -35.350
Time_sq        0.3768      0.005     83.425      0.000       0.368       0.386
==============================================================================
Omnibus:                      164.549   Durbin-Watson:                   0.090
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1611.955
Skew:                           1.969   Prob(JB):                         0.00
Kurtosis:                      13.474   Cond. No.                     1.27e+05
==============================================================================
```



- What can you say from the output? What about from the graph of the residuals?
- **In general we would like the residuals to be inside the interval in the graph.**

What mistake are we making?

What Looks Good (But Shouldn't Fool Us)

R^2 jumped from 0.865 to 0.994: Adding the quadratic term dramatically improved fit. The model now captures the curvature in GDP growth.

Durbin-Watson improved from 0.005 to 0.090: Still terrible (should be near 2), but better. Some of what looked like serial correlation was actually misspecified functional form.

Time_sq coefficient (0.377, $p < 0.001$): Highly significant, suggesting GDP growth has accelerated over time.

<u>The Mistake: This Is Still a Misspecified Model</u>

Look at the residual plot (green line). The residuals are not randomly bouncing around zero—they show clear patterns:

- Systematically negative in the 1950s-60s
- Positive in the 1970s-80s
- Negative around 2008-2010 (the financial crisis is visible!)
- Positive and growing recently

This is textbook evidence of serial correlation and remaining misspecification. The residuals should look like random noise if the model is correctly specified.

<u>The Deeper Problem</u>

You're modeling GDP with a **deterministic trend**—assuming GDP follows a smooth polynomial path with random deviations around it. But GDP has:

- **Business cycles** (expansions and recessions)
- **Structural breaks** (the 2008 crisis, COVID)
- **Stochastic trends** (shocks that permanently shift the level)

A quadratic trend can't capture these features. You're fitting a smooth curve through inherently cyclical, shock-driven data.

What Would Be Better?

This is likely setting up for AR models, differencing, or cointegration—approaches that model the *dynamics* of GDP rather than just its trend.

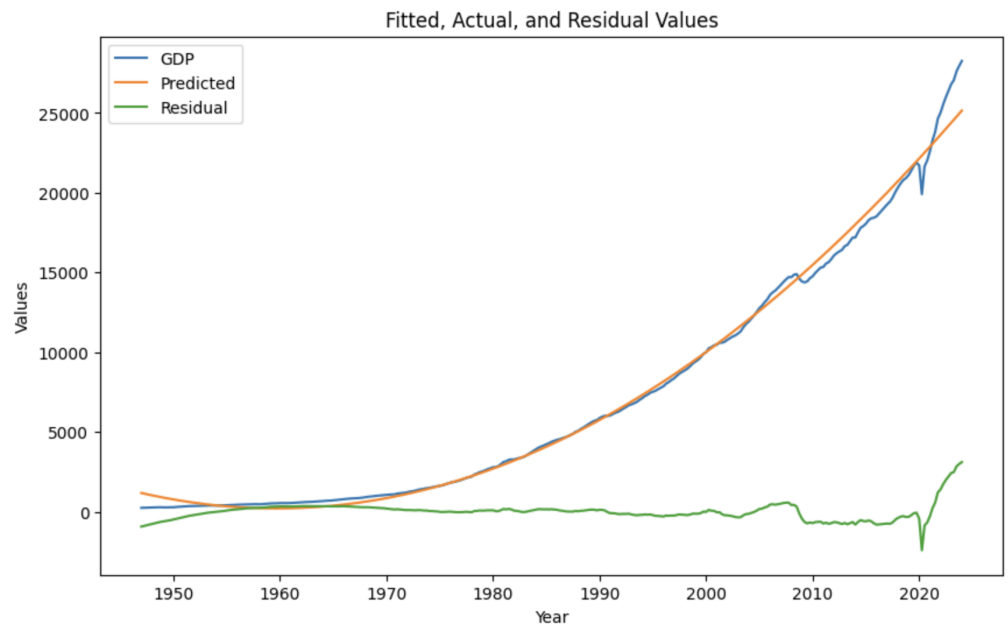
Is ?

Quadratic Trend Estimation?

Let us now try a quadratic deterministic trend model!

$$GDP_t = c + \beta_1 t + \beta_2 t^2 + u_t$$

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.994
Model:                            OLS   Adj. R-squared:                  0.994
Method:                 Least Squares   F-statistic:                 2.683e+04
Date:                Tue, 23 Jul 2024   Prob (F-statistic):               0.00
Time:                        20:19:05   Log-Likelihood:                -2395.0
No. Observations:                 309   AIC:                             4796.
Df Residuals:                     306   BIC:                             4807.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Constant    1174.2928     95.795     12.258      0.000     985.793    1362.792
Time         -38.1771      1.437    -26.570      0.000     -41.004     -35.350
Time_sq        0.3768      0.005     83.425      0.000       0.368       0.386
==============================================================================
Omnibus:                      164.549   Durbin-Watson:                   0.090
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1611.955
Skew:                           1.969   Prob(JB):                         0.00
Kurtosis:                      13.474   Cond. No.                     1.27e+05
==============================================================================
```

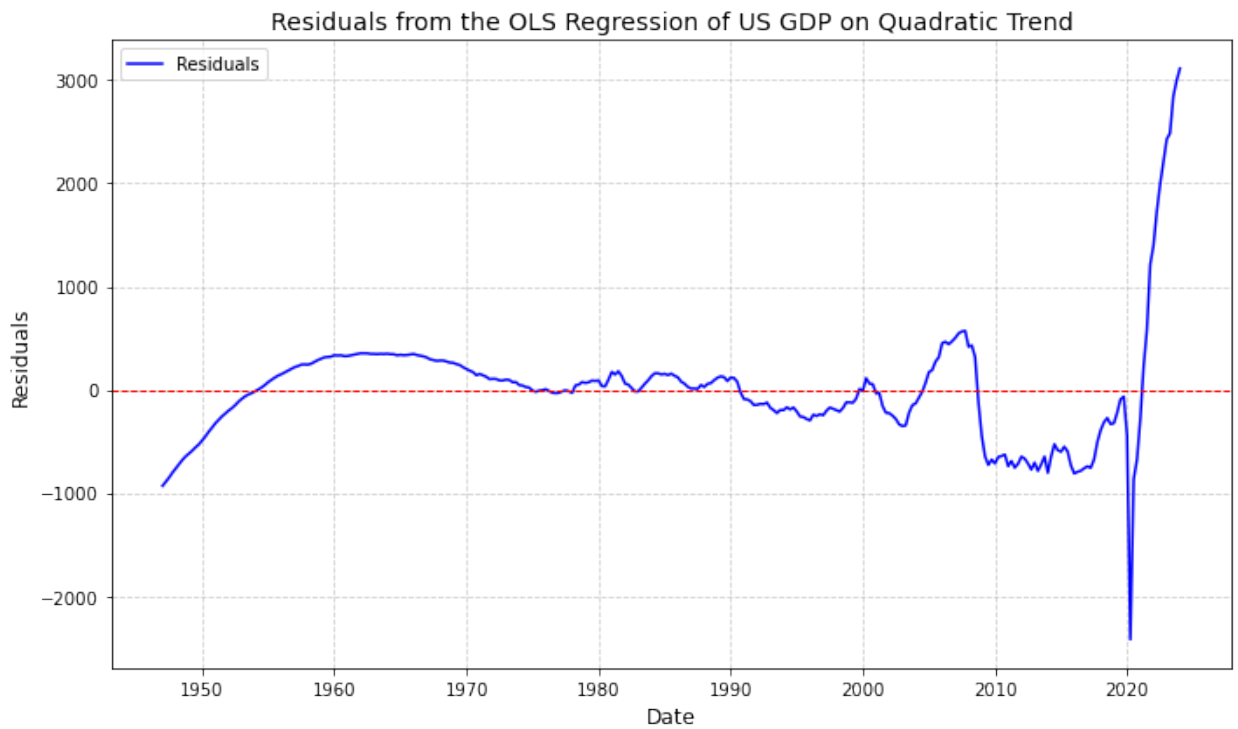


Deterministic Trends do not have an economics interpretation, yet it is incredibly powerful for forecasting.

Quadratic Trend

Let us now try a quadratic deterministic trend model!

$$GDP_t = c + \beta_1 t + \beta_2 t^2 + u_t$$

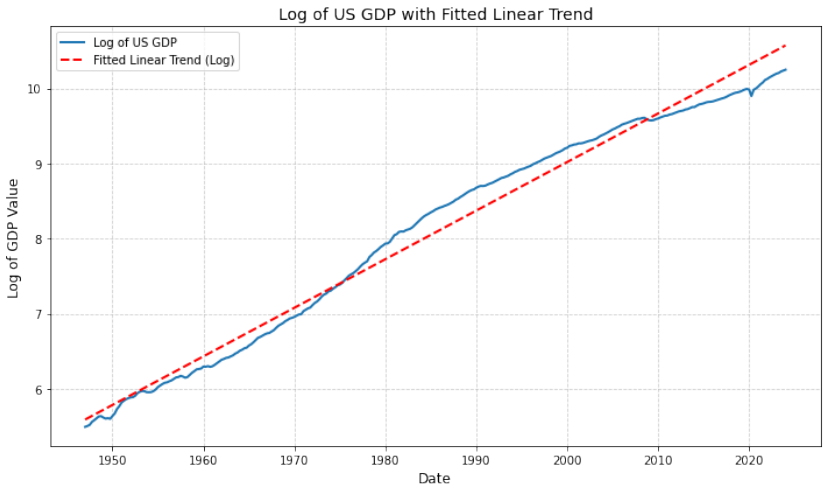


Log GDP with linear trend

Sometimes a log transformation helps. Let's look at logGDP (this is equivalent to say that USGDP has an exponential trend).

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.983
Model:                            OLS   Adj. R-squared:                  0.983
Method:                 Least Squares   F-statistic:                 1.753e+04
Date:                Sun, 05 Jan 2025   Prob (F-statistic):          7.25e-273
Time:                        10:52:13   Log-Likelihood:                 73.147
No. Observations:                 309   AIC:                            -142.3
Df Residuals:                     307   BIC:                            -134.8
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          5.5909      0.022    257.109      0.000       5.548       5.634
x1             0.0162      0.000    132.400      0.000       0.016       0.016
==============================================================================
Omnibus:                      151.864   Durbin-Watson:                   0.005
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               18.361
Skew:                           0.082   Prob(JB):                     0.000103
Kurtosis:                       1.817   Cond. No.                         355.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Why Log GDP Makes Sense

When you model $\log(GDP) = c + \beta t + u_t$, you're implicitly saying:

$GDP_t = e^{(c + \beta t + u_t)} = e^c \cdot e^{(\beta t)} \cdot e^{(u_t)}$

This is **exponential growth**—GDP grows by a constant *percentage* each period, not a constant *amount*. That's much more realistic economically. We talk about "3% growth," not "GDP increased by $500 billion."

The coefficient (0.0162): This is approximately the average growth rate per time period. If your data is quarterly, that's about 1.6% per quarter; if annual, 1.6% per year.

What Improved

The fit looks better visually. The linear trend through log GDP tracks the data more naturally than the quadratic through levels. The curvature you were trying to capture with t^2 is now handled by the log transformation.

$R^2 = 0.983$: Still very high, with a simpler (one regressor) model than the quadratic.

What's Still Wrong

Durbin-Watson = 0.005: We're right back to extreme serial correlation. The log transformation helped with functional form but didn't solve the autocorrelation problem.

Look at the graph—you can see the deviations from trend are persistent. The economy spends extended periods above or below trend (business cycles), and there are visible structural breaks (the dip around 2008-2009, the sharp COVID drop around 2020).

<u>The Fundamental Issue</u>

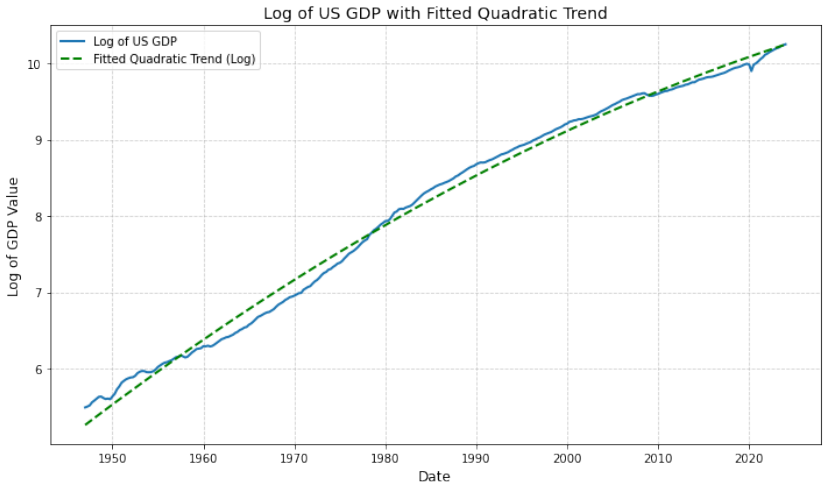
Whether you use levels, quadratics, or logs, a **deterministic trend model** assumes deviations from trend are temporary and independent. But macroeconomic shocks are persistent—that's why Durbin-Watson keeps telling you there's a problem.

This is likely motivating a move toward stochastic trend models, or at minimum using HAC standard errors. Is that coming next?

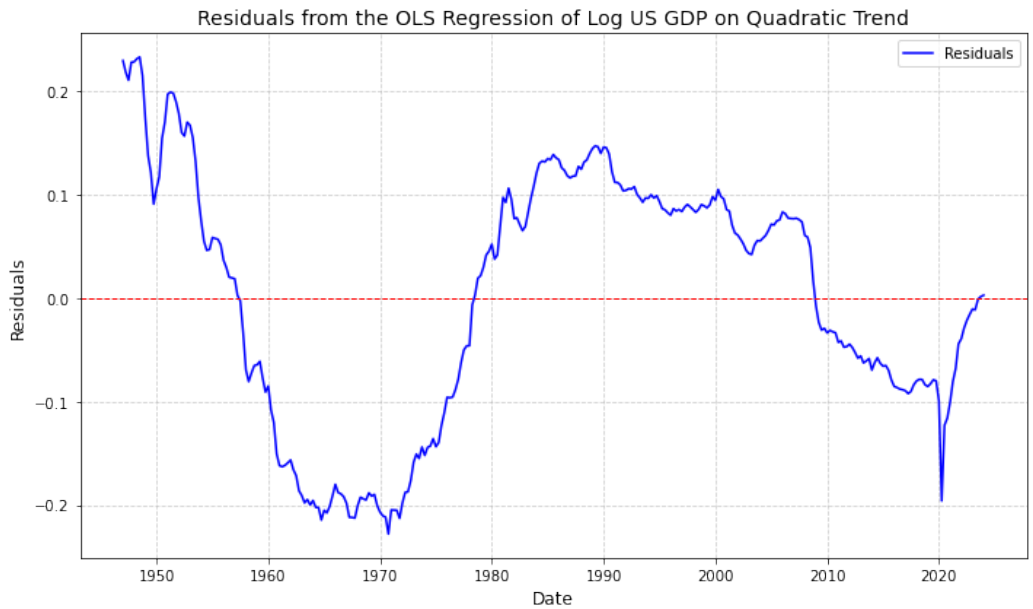
Log GDP with quadratic trend

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       0.993
Model:                            OLS   Adj. R-squared:                  0.993
Method:                 Least Squares   F-statistic:                 2.196e+04
Date:                Sun, 05 Jan 2025   Prob (F-statistic):               0.00
Time:                        11:04:52   Log-Likelihood:                 213.93
No. Observations:                 309   AIC:                            -421.9
Df Residuals:                     306   BIC:                            -410.7
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          5.2639      0.021    255.128      0.000       5.223       5.304
x1             0.0226      0.000     72.927      0.000       0.022       0.023
x2         -2.075e-05   9.73e-07    -21.333      0.000   -2.27e-05   -1.88e-05
==============================================================================
Omnibus:                       82.146   Durbin-Watson:                   0.011
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               17.856
Skew:                          -0.252   Prob(JB):                     0.000133
Kurtosis:                       1.936   Cond. No.                     1.27e+05
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.27e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
```



Log GDP with quadratic trend



This is showing the same pattern we saw before—adding complexity to chase fit, but not solving the underlying problem.

What Changed

Added t^2 to the log model: Now $\log(GDP_t) = c + \beta_1 t + \beta_2 t^2 + u_t$

The t^2 coefficient (-2.07e-05): Small but significant (t = -21.3). This negative sign suggests GDP growth has been *slowing down* over time—the growth rate in the 1950s-60s was higher than recently. That's actually economically plausible (mature economies tend to grow slower).

R^2 increased slightly (0.993 vs 0.983): Marginally better fit, as expected when adding a regressor.

The Same Fundamental Problem

Durbin-Watson = 0.011: Still essentially zero. We've gone from 0.005 (linear levels) → 0.090 (quadratic levels) → 0.005 (log linear) → 0.011 (log quadratic). None of these are anywhere close to 2.

The serial correlation won't go away because it's not a functional form problem—it's that the errors themselves are autocorrelated. Business cycles and persistent shocks don't disappear just because you transform the dependent variable or add polynomial terms.

The Warning Note at the Bottom

"The condition number is large, 1.27e+05. This might indicate strong multicollinearity."

This is because t and t^2 are highly correlated—when t is large, t^2 is also large. This inflates standard errors on individual coefficients, though the overall model predictions can still be okay.

Where This Is Heading

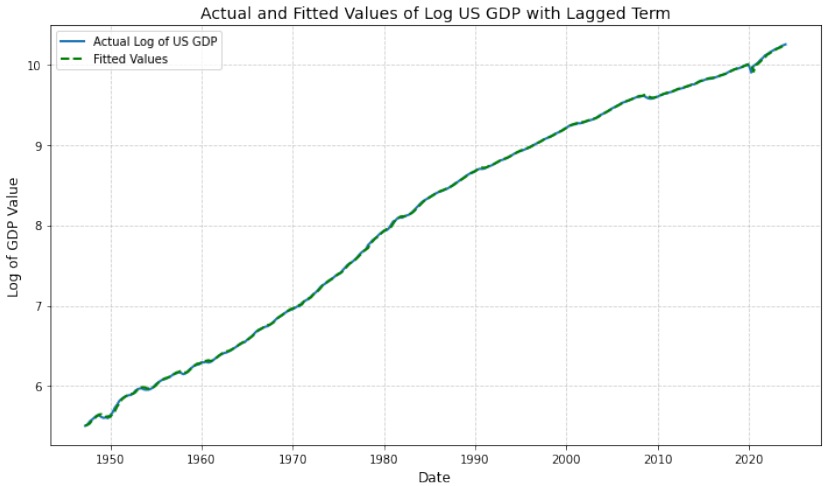
You've now tried four specifications, and Durbin-Watson has been terrible in all of them. The lesson: you need to model the **dynamics** explicitly—either with autoregressive terms, differencing (looking at GDP growth rates instead of levels), or using Newey-West standard errors to at least get valid inference.

Is the next slide introducing AR models or first-differencing?

Log GDP with quadratic trend and AR

Let's now add lagged USGDP in the regression. We will see later why this makes sense. Look also at what the previous graph suggest. What happens now?

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GDP   R-squared:                       1.000
Model:                            OLS   Adj. R-squared:                  1.000
Method:                 Least Squares   F-statistic:                 1.385e+06
Date:                Sun, 05 Jan 2025   Prob (F-statistic):               0.00
Time:                        11:21:51   Log-Likelihood:                 914.98
No. Observations:                 308   AIC:                            -1822.
Df Residuals:                     304   BIC:                            -1807.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0750      0.031      2.434      0.015       0.014       0.136
x1             0.9887      0.006    168.539      0.000       0.977       1.000
x2             0.0003      0.000      2.209      0.028    3.29e-05       0.001
x3         -4.697e-07   1.58e-07     -2.974      0.003    -7.8e-07   -1.59e-07
==============================================================================
Omnibus:                      108.259   Durbin-Watson:                   1.558
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3036.847
Skew:                          -0.772   Prob(JB):                         0.00
Kurtosis:                      18.305   Cond. No.                     1.88e+06
==============================================================================
```



Durbin-Watson = 1.558 — now we're in reasonable territory! Let me break down what this model is telling us.

The Durbin-Watson Improvement

We went from ~0.005-0.011 (extreme positive serial correlation) to 1.558. While 2.0 would indicate no autocorrelation, 1.558 is vastly better—likely in the "inconclusive" or "acceptable" range depending on your critical values. The AR term absorbed most of the persistence in the errors.

Interpreting the Coefficients

The model appears to be: $\log(GDP_t) = c + \beta_1 \cdot \log(GDP_{t-1}) + \beta_2 \cdot t + \beta_3 \cdot t^2 + u_t$

x1 (lagged log GDP) = 0.9887: This is the key coefficient. GDP is highly persistent—knowing last period's GDP tells you almost everything about this period's GDP. A coefficient this close to 1 suggests GDP may have a **unit root** (stochastic trend), which has important implications for inference and forecasting.

x2 (time) = 0.0003: A small positive linear trend component, conditional on the lagged term . trend

x3 (time squared) = -4.697e-07: Tiny negative quadratic term, suggesting slight deceleration in trend growth. quadratic trend

R^2 = 1.000?

This looks suspicious but isn't wrong—with a lagged dependent variable that's 0.99 correlated with the current value, you're explaining almost all variation mechanically. This is why R^2 is less meaningful in AR models.

The Bigger Picture

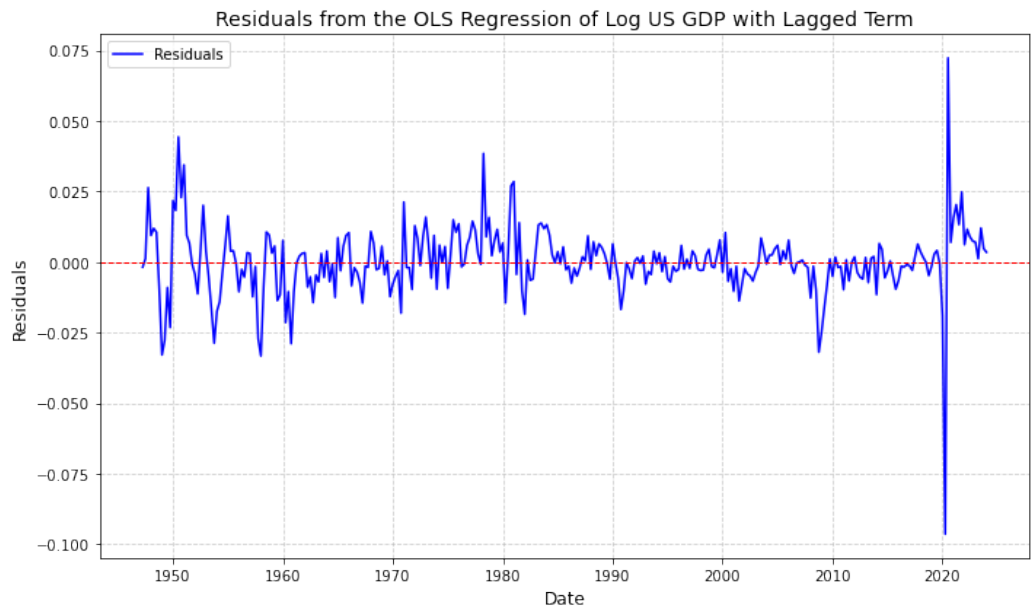
What the previous residual graph suggested: The residuals showed volatility clustering (larger errors during crises) and potential structural breaks. Adding the AR term helped with serial correlation but won't fix heteroskedasticity or outliers from COVID/2008.

The near-unit-root issue: With $\beta_1 = 0.9887$, you're close to a random walk. This raises questions about whether you should be modeling GDP in levels at all, or whether first-differencing (modeling GDP *growth*) would be more appropriate.

Is the course heading toward unit root tests next?

Log GDP with quadratic trend and AR- residuals

Look at the residuals now. Look carefully and discuss! There is a lot going on here.



$$Y_t = \alpha + B_t + \gamma y_{t-1} + u_t$$

$\downarrow$

log GDP

	y_t	y_{t-1}
1948 : 1	6.5	
1948 : 2	7.3	6.5
1948 : 3	3.4	7.3
	⋮	

Now we're getting somewhere. Adding an AR term (lagged dependent variable) has dramatically changed the residual picture. Let me walk through what I see.

The Good News

Residuals now look much more like white noise. Compare this to the earlier residual plots—those had obvious trends and persistent deviations. Here, the residuals bounce around zero much more randomly throughout most of the sample. This is what we *want* residuals to look like.

The serial correlation problem appears largely solved by including the lagged term. When you model $y_t = \rho y_{t-1} + X\beta + u_t$, you're explicitly capturing the persistence that was showing up in the Durbin-Watson statistic.

The Things to Discuss

1. Volatility clustering in recent years (2020+): Look at the right side of the graph—the residuals get much larger and more volatile. That's COVID. The pandemic caused GDP movements that were completely outside historical norms. This is a structural break, and no time trend model will capture it well.

2. The 2008-2009 period: There's also visible disruption around the financial crisis—larger residuals than typical.

3. Pre-1960 vs. post-1960: The earlier period looks slightly more volatile. This could be real (the "Great Moderation" hypothesis—that macro volatility decreased after the 1980s) or it could be data quality issues with older GDP measurements.

4. Heteroskedasticity: The variance of residuals isn't constant over time—it's larger during crisis periods. This suggests homoskedasticity (Assumption 5) is violated, which would call for robust standard errors even with the AR correction.

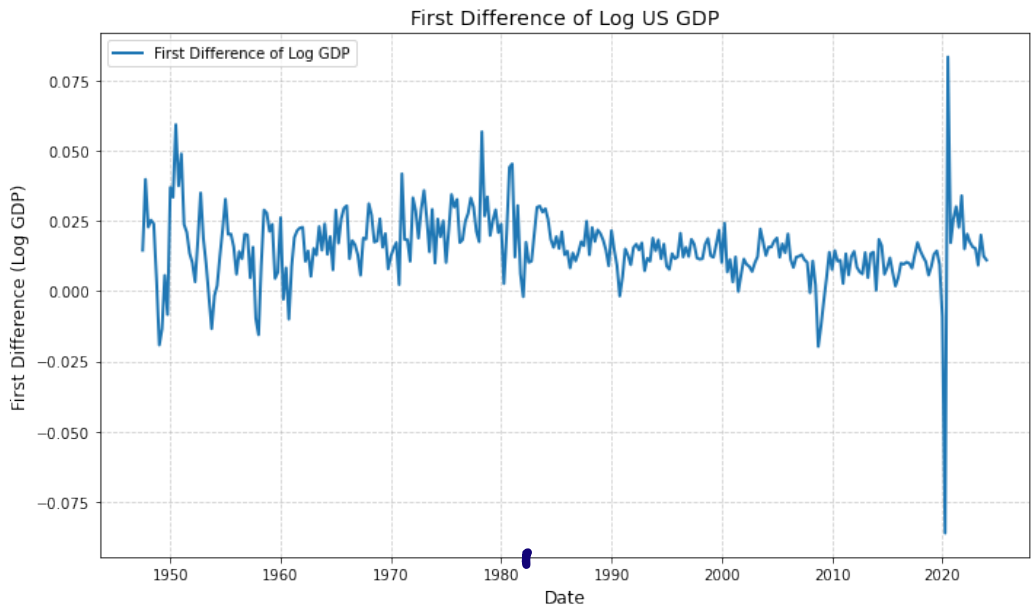
The Broader Lesson

Adding dynamics (the AR term) addressed the serial correlation that polynomial trends couldn't fix. But extreme events (financial crises, pandemics) will always show up as outliers—no standard model handles those gracefully.

What does the Durbin-Watson look like now with the AR term included?

GDP growth

As a side note. Look at how GDP growth ($log(y_t) - log(y_{t-1})$ looks like. Discuss. More on this later in the semester...



198

Model Selection

* The better the model, the better the forecasting

R-Square

- You may have notice that R-Square in our models is very high.
- Once you add a trend or a lag the R-Square quicky get's high.
- Typical of macro data. Finance is a little different.

Output Analysis

- Still look at R-Square and Adjusted R-Square
- Plot the residuals
- Careful with SE if you have not modeled all the dependency
- Information Criteria are the ideal way to compare models especially when they are nested.
- Durbin-Watson statistics

nested model: one model inside the other. Equal to one model when a variable is equal to zero

Information Criteria

How do we select between different models?

- The model with the highest R^2 is not always the best model for out of sample forecast.

AIC :

BIC :

! when you have log, AIC + BIC will be negative.

What They're Doing

Both AIC and BIC are trying to solve the same problem: R^2 always increases when you add variables, so you need some way to penalize model complexity. They balance fit against parsimony.

AIC (Akaike Information Criterion):
AIC = -2·ln(L) + 2k

BIC (Bayesian Information Criterion):
BIC = -2·ln(L) + k·ln(n)

Where L is the likelihood, k is the number of parameters, and n is the sample size.

The Key Difference

BIC penalizes complexity more heavily when n > 7 (since ln(n) > 2 for n > 7). With your n = 308, the BIC penalty is ln(308) ≈ 5.7 per parameter, versus AIC's penalty of 2 per parameter.

Practical implications:

- AIC tends to choose larger models (more variables)
- BIC tends to choose smaller models (fewer variables)
- BIC is consistent (will select the true model asymptotically if it's in your candidate set)
- AIC is efficient (minimizes prediction error)

Looking at Your Output
From the AR model slide:
- AIC = -1822
- BIC = -1807

Lower is better for both. When comparing models, you'd pick the one with the lower AIC (or BIC). If AIC and BIC disagree about which model is best, it usually means AIC is favoring a more complex model and BIC is favoring a simpler one.

When to Use Which?

Use AIC if your goal is prediction—you want the best forecasting model.
Use BIC if your goal is identifying the "true" model structure—you want to avoid overfitting and select the most parsimonious specification.

In time series contexts like this GDP example, BIC is often preferred for selecting lag length in AR models

Information Criteria

How do we select between different models?

- The model with the highest R^2 is not always the best model for out of sample forecast.
- Most ICs attempt to find the model with the smallest out-of-sample 1-step-ahead mean squared error.

$$MSE = \frac{\sum_{t=1}^{T} e_t^2}{T} \quad \text{where } e_t = y_t - \hat{y}_t = \hat{u}_t$$

R^2 and Information Criteria

Note that

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum e_t^2}{\sum (y_t - \bar{y})^2}$$

Where TTS depends only on the data.

Smallest MSE $\Leftrightarrow$ highest R^2

We could look at the model with smallest MSE. We said that R^2 increases even if we include irrelevant variables so that we needed to correct for the degrees of freedom (d.f.). The same is true for MSE.

Information Criteria

Different Information Criteria correct MSE for the degrees of freedom with different weights.

Information Criteria

Different Information Criteria correct MSE for the degrees of freedom with different weights.

- Akaike Information Criteria (AIC)

$$AIC = e^{\frac{2k}{T}} MSE$$

- Schwartz Information Criteria (SIC) → same as BIC

$$SIC = T^{\frac{k}{T}} MSE$$

That's why you look at k relalative to T

degrees of freedom?

(N-k) (T-k)

I want this number to be high, if it's too small you run into trouble

The more data you have, the more degrees of freedom, you can fit a better line.

Information Criteria

Different Information Criteria correct MSE for the degrees of freedom with different weights.

- Akaike Information Criteria (AIC)

$$AIC = e^{\frac{2k}{T}} MSE$$

- Schwartz Information Criteria (SIC)

$$SIC = T^{\frac{k}{T}} MSE$$

Between two models, pick the one with the smallest <u>*AIC and SIC*</u>*.Carefull with logged data (smallest in the real line)*

Durbin Watson Test — not that much line?

- The Durbin Watson test is one way to test to see if you still have serial correlation in the residuals.
- $\varepsilon_t = \phi\varepsilon_{t-1} + v_t$, the DW statistics tests the null that $\phi = 0$
- The formula for the test is:

$$DW = \frac{\sum_{t=2}^{T} (\varepsilon_t - \varepsilon_{t-1})^2}{\sum_{t=1}^{T} \varepsilon_t^2}$$

- DW takes values in the interval $[0, 4]$, and if all is well, DW should be around 2.
- If DW is substantially less than 2 , there is evidence of positive serial correlation. A rough rule of thumb, if DW is less than 1.5.
- For log GDP, DW was 0.011 with quadratic trend and 1.558 once we added the lag.